

Business Requirements Document (BRD)

Algorithmic Trading System

Table of Contents

- [Business Requirements Document \(BRD\)](#)
 - [Algorithmic Trading System](#)
 - [Table of Contents](#)
 - [Executive Summary](#)
 - [Introduction](#)
 - [Business Objectives](#)
 - [Scope](#)
 - [Stakeholders](#)
 - [Functional Requirements](#)
 - [User Management](#)
 - [Core Trading Infrastructure](#)
 - [Data Management and Analytics](#)
 - [AI-Powered Strategy Development and Backtesting](#)
 - [Agentic AI Assistant](#)
 - [User Experience and Multi-Platform Accessibility](#)
 - [Enterprise-Grade Security, Compliance, and Operations](#)
 - [Options Trading](#)
 - [Non-Functional Requirements](#)
 - [Constraints](#)
 - [Assumptions](#)
 - [Risks](#)
 - [Glossary](#)

Executive Summary

The Algorithmic Trading System is an innovative, enterprise-grade platform designed to revolutionize the trading landscape by democratizing access to advanced trading tools and strategies. This system integrates a sophisticated core trading engine, AI-powered tools, an Agentic AI Assistant, and a unified user interface to empower traders of all experience levels—from novices to professionals—to navigate financial markets with precision and confidence. By leveraging a "Best-of-Breed" integration strategy with leading open-source technologies, the platform ensures modularity, scalability, and cutting-edge functionality.

This Business Requirements Document (BRD) outlines the business objectives, scope, stakeholders, and detailed functional and non-functional requirements necessary to develop this transformative system. The primary goals include democratizing trading tools, driving innovation through artificial intelligence, ensuring flexibility and scalability, and empowering users with actionable insights and automation. The scope encompasses multi-asset trading capabilities, advanced data analytics, AI-driven strategy development, natural language interaction via the AI Assistant, a responsive multi-platform interface, robust security measures, and comprehensive options trading features.

Key stakeholders include traders, portfolio managers, compliance officers, developers, project managers, and business analysts, each with distinct roles in ensuring the system's success. This BRD provides a clear roadmap for development, aligning business needs with technical implementation, and is tailored to support the use of Kilo Code, an AI Coding App, to streamline the coding process.

Introduction

The Algorithmic Trading System is a next-generation platform engineered to bridge the gap between sophisticated financial technologies and practical trading applications. By integrating advanced trading tools, artificial intelligence, and an intuitive interface, the system aims to make complex trading strategies accessible to a diverse user base. The platform adopts a microservices architecture, leveraging best-in-class open-source technologies to deliver a modular, scalable, and high-performance solution.

This BRD serves as the cornerstone for defining the business needs and requirements, providing a comprehensive guide for stakeholders and

developers. It is designed to work in tandem with supporting documents like the Product Requirements Document (PRD), Technical Specification Document (TSD), API Documentation, Test Plan, and Features & Phases, ensuring a holistic approach to system development.

Business Objectives

The Algorithmic Trading System is driven by the following business objectives:

- **Democratization of Trading Tools:** Provide access to advanced trading strategies and technologies traditionally reserved for institutional investors, leveling the playing field for retail and professional traders alike.
- **Innovation through AI:** Harness artificial intelligence and machine learning to deliver intuitive, powerful, and user-friendly trading capabilities, enhancing decision-making and automation.
- **Flexibility and Scalability:** Create a platform that adapts to the evolving financial landscape, supports a growing user base, and accommodates new features and asset classes over time.
- **User Empowerment:** Bridge cutting-edge technology with practical trading applications, equipping users with the tools and insights needed to succeed in dynamic markets.

These objectives align with the vision of transforming algorithmic trading and setting new industry standards for accessibility, performance, and innovation.

Scope

The Algorithmic Trading System encompasses the following components and capabilities:

- **Core Trading Engine:** Supports trading across multiple asset classes (stocks, ETFs, futures, options, forex, cryptocurrencies) with integrations to various brokers.
- **Data Management and Analytics:** Provides real-time and historical market data, alternative data integration, and advanced analytics for informed decision-making.

- **AI-Powered Tools:** Enables strategy creation, backtesting, and optimization through no-code and code-based interfaces, enhanced by generative AI.
- **Agentic AI Assistant:** Offers natural language interaction for trading commands, portfolio insights, and document-based queries, powered by a collaborative agent network.
- **Unified User Interface:** Delivers a responsive, multi-platform experience (web and mobile) with personalized dashboards and institutional-grade charting.
- **Enterprise-Grade Security and Compliance:** Ensures robust security, regulatory compliance, and risk management features suitable for enterprise use.
- **Options Trading:** Provides comprehensive tools for pricing, analyzing, and trading options, including complex strategies and risk analytics.

Out of Scope:

- Development of third-party broker systems beyond defined integration points.
- Management of external data sources beyond fallback mechanisms and specified integrations.
- Hardware infrastructure provisioning, which is assumed to be handled by deployment teams.

Stakeholders

The following stakeholders are critical to the development and success of the Algorithmic Trading System:

- **Traders:** Primary users, ranging from novices seeking simplicity to professionals requiring advanced tools.
- **Portfolio Managers:** Users managing multiple accounts or large portfolios, needing real-time insights and risk management.
- **Compliance Officers:** Ensure the system adheres to regulatory standards and provides audit capabilities.
- **Developers:** Build and maintain the system, translating requirements into functional code.

- **Project Managers:** Oversee development timelines, resource allocation, and delivery milestones.
- **Business Analysts:** Define requirements, validate alignment with business goals, and facilitate stakeholder communication.

Functional Requirements

The system must deliver the following functionalities, organized by major components:

User Management

- **FR-UM-01:** The system shall allow users to register, log in, and manage their profiles via a secure interface.
- **FR-UM-02:** The system shall support multiple user roles (e.g., Admin, Trader, Quant, Risk Manager) with role-based access control (RBAC).
 - *Acceptance Criteria:* Users can only access features permitted by their role (e.g., Traders can place orders, Admins can manage users).
- **FR-UM-03:** The system shall authenticate users using JSON Web Tokens (JWT) and support Single Sign-On (SSO) with major identity providers.
- **FR-UM-04:** The system shall provide user preference settings (e.g., language, dashboard layout) saved to their profile.

Core Trading Infrastructure

- **FR-CTI-01:** The system shall integrate with multiple brokers, including Interactive Brokers, Alpaca, and TD Ameritrade, for trade execution.
 - *Details:* Initial integration with Interactive Brokers, expandable to others in later phases.
- **FR-CTI-02:** The system shall support trading across asset classes: stocks, ETFs, futures, options, forex, and cryptocurrencies.
 - *Acceptance Criteria:* Users can trade any supported asset class via the interface or API.
- **FR-CTI-03:** The system shall provide an Order Management System (OMS) supporting:
 - Basic order types: Market, Limit, Stop, Stop-Limit, Trailing Stop.
 - Advanced order types: Good-Till-Canceled (GTC), Immediate-or-Cancel (IOC), Fill-or-Kill (FOK), Volume-Weighted Average Price (VWAP), Time-

Weighted Average Price (TWAP), Iceberg, One-Cancels-the-Other (OCO).

- *Acceptance Criteria:* All order types execute correctly under normal market conditions.
- **FR-CTI-04:** The system shall log all order executions and provide error handling for failed orders (e.g., insufficient funds, connectivity issues).
- **FR-CTI-05:** The system shall enable seamless switching between paper and live trading modes without requiring manual reconfiguration.

Data Management and Analytics

- **FR-DMA-01:** The system shall integrate with free data sources (e.g., Yahoo Finance, Alpha Vantage) for paper trading and Interactive Brokers for live trading.
- **FR-DMA-02:** The system shall implement a fallback mechanism for data sources, switching to alternatives if the primary source fails.
 - *Details:* Order of fallback per asset class as specified in API Documentation (e.g., Stocks: IB → Yahoo → Alpha Vantage).
- **FR-DMA-03:** The system shall store and provide access to historical market data with a minimum depth of 5 years for backtesting and analysis.
- **FR-DMA-04:** The system shall integrate alternative data sources, including real-time news feeds with sentiment analysis, fundamental data, and economic indicators.
- **FR-DMA-05:** The system shall offer market analytics, including:
 - Advanced market screener with natural language filters.
 - AI-powered opportunity detection across asset classes.
 - Market microstructure analysis (e.g., order book dynamics).
- **FR-DMA-06:** The system shall provide post-trade analytics, including Transaction Cost Analysis (TCA), fill rate analysis, and broker performance evaluation.

AI-Powered Strategy Development and Backtesting

- **FR-AIS-01:** The system shall provide a no-code strategy builder with a drag-and-drop interface for creating trading strategies.
 - *User Story:* As a novice trader, I want to build a moving average crossover strategy without coding.

- **FR-AIS-02:** The system shall include a generative AI tool to convert natural language prompts into executable strategy code.
 - *Example:* "Create a strategy that buys when the 50-day SMA crosses above the 200-day SMA."
- **FR-AIS-03:** The system shall offer a Python-based studio with cloud-hosted Jupyter notebooks for advanced strategy development.
- **FR-AIS-04:** The system shall provide an enterprise-grade backtesting engine with:
 - Realistic modeling (slippage, commissions, market impact).
 - Advanced analytics (Monte Carlo stress testing, walk-forward optimization).
 - AI-assisted optimization and overfitting detection.
- **FR-AIS-05:** The system shall include a strategy marketplace for users to share and license trading strategies.

Agentic AI Assistant

- **FR-AIA-01:** The system shall provide an AI Assistant capable of understanding and responding to natural language queries related to trading and portfolio management.
 - *Examples:* "What's my portfolio value?" or "Place a limit order for 10 shares of AAPL at \$150."
- **FR-AIA-02:** The AI Assistant shall execute trading commands based on user instructions, integrating with the OMS.
- **FR-AIA-03:** The AI Assistant shall process user-uploaded documents (e.g., PDFs, spreadsheets) and answer questions based on their content using Agentic Retrieval-Augmented Generation (RAG).
- **FR-AIA-04:** The system shall implement a collaborative agent network with specialized roles (e.g., Analyst, Researcher, Risk Manager, Compliance) to delegate tasks.
- **FR-AIA-05:** The AI Assistant shall support continuous learning, incorporating user feedback to improve accuracy over time.

User Experience and Multi-Platform Accessibility

- **FR-UE-01:** The system shall provide a unified, responsive web interface with personalized dashboards and drag-and-drop widgets.

- **FR-UE-02:** The interface shall integrate institutional-grade charting (e.g., TradingView) for real-time and historical data visualization.
- **FR-UE-03:** The system shall include a conversational chatbot interface for interacting with the AI Assistant, supporting text, file uploads, and voice commands.
- **FR-UE-04:** The system shall support multi-platform access:
 - Web application built with Next.js, responsive across devices.
 - Progressive Web App (PWA) with offline capabilities.
 - Future native mobile apps for iOS and Android using React Native.
- **FR-UE-05:** The system shall comply with WCAG 2.1 accessibility standards and support multi-language interfaces with real-time translation.

Enterprise-Grade Security, Compliance, and Operations

- **FR-SEC-01:** The system shall implement a Zero-Trust Architecture with continuous verification of all requests.
- **FR-SEC-02:** The system shall comply with regulatory standards (e.g., GDPR, HIPAA, financial regulations) and provide automated compliance reporting and immutable audit trails.
- **FR-SEC-03:** The system shall include a real-time risk management hub with:
 - Pre-trade risk checks (e.g., concentration risk, margin limits).
 - Circuit breakers to halt trading under predefined conditions.
 - Advanced risk analytics (e.g., Value at Risk, correlation analysis).
- **FR-SEC-04:** The system shall support operational monitoring with Prometheus and Grafana for metrics visualization and alerting.
- **FR-SEC-05:** The system shall provide user management features, including RBAC and SSO integration.

Options Trading

- **FR-OPT-01:** The system shall support trading of basic, complex, and exotic options strategies (e.g., Iron Condors, Butterflies).
- **FR-OPT-02:** The system shall provide tools for:
 - Calculating Greeks (Delta, Gamma, Theta, Vega).
 - Generating payoff diagrams and pricing models (e.g., Heston, Monte Carlo).

- **FR-OPT-03:** The system shall include a volatility suite with surface constructors and skew arbitrage detectors.
- **FR-OPT-04:** The system shall offer risk management tools specific to options, such as Greek exposure analysis and margin forecasting.

Non-Functional Requirements

The system must meet the following non-functional standards:

- **NFR-PERF-01: Performance**
 - Order execution latency shall be under 100ms under normal conditions.
 - API response times shall be below 100ms for critical operations.
- **NFR-SCALE-02: Scalability**
 - The system shall support horizontal scaling to handle a growing user base and data volume.
 - Independent scaling of components (e.g., trading engine, AI service) shall be possible.
- **NFR-SEC-03: Security**
 - The system shall comply with OWASP Top 10 security standards.
 - Regular security audits and penetration testing shall be conducted.
- **NFR-REL-04: Reliability**
 - The system shall achieve 99.99% uptime with robust failover mechanisms.
 - Automated backups and disaster recovery plans shall be implemented.
- **NFR-USAB-05: Usability**
 - The interface shall be intuitive with a minimal learning curve for new users.
 - Comprehensive documentation and user guides shall be provided.
- **NFR-COMP-06: Compliance**
 - The system shall adhere to financial regulations and data protection laws (e.g., GDPR).
 - Tools for automated compliance reporting shall be available.

Constraints

- **CON-01:** The system must leverage specified open-source technologies (e.g., QuantConnect LEAN, LangChain, Next.js, QuantLib) as outlined in the TSD and Features & Phases.
- **CON-02:** Development must adhere to the 5-phase roadmap detailed in the Features & Phases document.
- **CON-03:** The system must comply with all relevant financial regulations and data protection laws, as specified in the PRD and Test Plan.

Assumptions

- **ASM-01:** Users possess basic knowledge of trading concepts and financial markets.
- **ASM-02:** The system will have access to necessary market data feeds and broker APIs as specified in the API Documentation.
- **ASM-03:** Sufficient and relevant data will be available to train AI models for accurate performance.
- **ASM-04:** Infrastructure (e.g., cloud hosting, network connectivity) will be provided to support deployment.

Risks

The following risks and mitigation strategies are identified:

- **RSK-01: Data Source Failures**
 - *Impact:* Disruption of market data availability.
 - *Mitigation:* Implement fallback mechanisms as per the API Documentation (e.g., switching to alternative sources).
- **RSK-02: Performance Bottlenecks**
 - *Impact:* Slow order execution or system lag.
 - *Mitigation:* Conduct early load testing and optimize critical paths (e.g., caching, asynchronous processing).
- **RSK-03: Security Vulnerabilities**
 - *Impact:* Unauthorized access or data breaches.
 - *Mitigation:* Perform regular security audits, adhere to Zero-Trust principles, and use encryption standards.

- **RSK-04: AI Misinterpretations**
 - *Impact*: Incorrect trading actions or responses.
 - *Mitigation*: Use diverse test cases, incorporate user feedback, and refine AI models continuously.

Glossary

- **Agentic RAG**: Retrieval-Augmented Generation system with AI agents for enhanced information processing and task delegation.
- **Zero-Trust Architecture**: Security model requiring continuous verification of all requests, assuming no inherent trust.
- **OMS**: Order Management System for placing, tracking, and managing trading orders.
- **Greeks**: Metrics (e.g., Delta, Gamma) used to assess options risk and pricing.
- **LEAN**: QuantConnect's algorithmic trading engine, forming the core trading infrastructure.

This Business Requirements Document (BRD) provides a detailed and elaborate specification for the Algorithmic Trading System, capturing all essential business needs and requirements. It is structured to support development with Kilo Code, ensuring clarity and precision for AI-assisted coding. By integrating insights from the provided documents, this BRD ensures alignment between business goals and technical implementation, paving the way for a robust, innovative trading platform.